Analysing University Students' Monthly Expenditure Using Robust Regression

Ann Abigail Halim
Computer Science and Statistics
BINUS University
Jakarta, Indonesia
ann.halim@binus.ac.id

Carmenita Angelica
Computer Science and Statistics
BINUS University
Jakarta, Indonesia
carmenita.angelica@binus.ac.id

Joycelin
Computer Science and Statistics
BINUS University
Jakarta, Indonesia
joycelin@binus.ac.id

***Abstract*—Rising living expenses put university students under greater financial strain, affecting their spending habits and overall well-being. This study uses data from an online survey conducted at BINUS University to examine the behavioural and financial factors influencing undergraduate students' monthly spending. The dataset includes 45 valid responses with details on living arrangements, financial behaviours, consumption patterns, and income capacity. To identify essential factors, the relationship between total monthly expenditure and ten explanatory variables was first examined using a multiple linear regression with Ordinary Least Squares (OLS). The presence of outliers and high-leverage observations, as well as violations of the normality assumption, was identified using classical regression diagnostics, suggesting possible sensitivity of the OLS estimator. To address these issues, a robust regression model based on the MM-estimator was subsequently implemented.**

Keywords—student expenditure, robust regression, ordinary least squares, regression diagnostics, financial behaviour, R programming

I. Introduction

Nowadays, university students face increasing pressure in financial management. Basic need insecurities are a prevalent and growing issue among university students, motivated by current rises in living costs [1]. This comes in addition to poor financial decisions that can lead to deteriorating physical and mental health [2], [3]. This can negatively impact the process by which these students are studying [7].

The financial strain caused by these issues is a concerning public health issue [3]. Demonstrated by a national survey of Norwegian students, finding that students with financial problems are more likely to experience mental health issues, such as anxiety and depression, as well as physical symptoms like headache and shoulder pain [4]. This results in a lower quality of life, sometimes leading to poorer academic performance and even worse results [7].

Prior research has built an understanding of university students' financial lives and their concerns. Wan Nawang et al. (2025) explored the determinants of overall Financial Well Being (FWB) [5]. Others focused on identifying key influencers, such as income, financial literacy, and spending habits among students [6]. Based on this, a significant gap in methodological rigour can be identified, such as the susceptibilities of outliers or non-normal distributions [8], [10].

This study aims to provide a methodologically sound analysis by ensuring the reliability of the findings. The purpose of this research is to assess whether the multiple linear regression (OLS) model satisfies the classical assumptions required for valid estimation. To evaluate whether the robust regression model provides more stable and reliable parameter estimates compared to the OLS model. To identify the financial and behavioural factors that significantly influence university students' monthly expenditures.

A. Objectives

1. To assess whether the multiple linear regression model satisfies the classical assumptions required for valid estimation.
2. To evaluate whether the robust regression model provides more stable and reliable parameter estimates compared to the multiple linear regression model.
3. To identify the financial and behavioural factors that significantly influence university students' monthly expenditures.

II. Multiple Linear Regression

A. Ordinary Least Square (OLS) in Multiple Linear Regression

Ordinary Least Squares (OLS) is the most widely used estimation method in linear regression, aiming to estimate regression coefficients by minimising the sum of squared residuals between observed and predicted values [11]. OLS provides closed-form solutions and optimal properties under classical regression assumptions, making it a standard baseline method in statistical modelling [11], [12].

B. Multiple Linear Regression Using OLS Formulation

Consider a multiple linear regression model.

$$Y = X\beta + \varepsilon,$$

where Y is the response vector, X is the design matrix, β is the vector of regression coefficients, and ε denotes the error term [12].

The OLS estimator of β is obtained by

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Model fit is typically assessed using the sum of squares decomposition:

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$$SST = \sum(y_i - \bar{y})^2, SSR = \sum(\hat{y}_i - \bar{y})^2, SSE = \sum(y_i - \hat{y}_i)^2,$$

with SST = SSR + SSE[12].

C. Assumption of Multiple Linear Regression

Multiple Linear Regression using the OLS method relies on several classical assumptions: the absence of perfect or severe linear correlation among the independent variables (non-multicollinearity), independence of the error terms (no autocorrelation), constant error variance (homoscedasticity), and normality of the errors for valid statistical inference [11], [12]. These assumptions ensure unbiased estimation, minimum variance, and reliable hypothesis testing [2].

1) Multicollinearity (Non-multicollinearity)

This test ensures that the independent variables are not so highly correlated that they mask each other's individual effects.

Hypothesis:

H_0: There is no significant multicollinearity among the predictors.

H_1: There is a significant presence of multicollinearity.

Reject H_0, if the Variance Inflation Factor (VIF) > 10 [11].

2) Autocorrelation (Independence of Errors)

This assumption is critical for ensuring that the residuals are independent of one another, which is a requirement for the OLS estimator to be efficient.

$H_0: \rho = 0$, The error terms are independent (n-autocorrelated).

$H_1: \rho \neq 0$, The error terms are correlated (autocorrelated).

Reject H_0, if $0 < DW < d_L$ or $4 - d_L < DW < 4$ [13].

3) Homoscedasticity (Constant Error Variance)

This ensures that the variance of the error term remains constant across all levels of the independent variables.

Hypothesis:

$H_0: Var\ (u|x) = \sigma^2$, The errors are homoscedastic (variance is constant).

$H_1: Var\ (u|x) \neq \sigma^2$, The errors are heteroscedastic (variance is non-constant)

Reject H_0, if $p - value < \alpha$ (0.05) using Breusch-Pagan [13].

4) Normality of Errors

This assumption is necessary for the validity of the t-tests and F-tests used to determine the significance of the coefficients.

Hypothesis:

H_0: Residuals are normally distributed.

H_1: Residuals are not normally distributed.

Reject H_0, if $p - value < \alpha$ (0.05) using tests such as Jarque-Bera, Shapiro-Wilk, or Kolmogorov-Smirnov (Lilliefors Test) [11], [13].

D. Limitations of the OLS Method

In practical applications, regression assumptions are often violated due to outliers, skewed distributions, measurement errors, or heterogeneous variance structures [14]. These violations can result in inefficient parameter estimates and misleading inference when OLS is used [14].

E. Model Evaluation

Model performance is commonly evaluated using the coefficient of determination R^2, which measures the proportion of variance explained by the model [11], [12]. Residual diagnostics, such as residual plots and normal probability plots, are frequently employed to assess the validity of model assumptions [12], [14].

III. Robust Regression

A. Definition of Robust Regression

Robust regression encompasses methods that yield reliable parameter estimates when data deviate from classical OLS assumptions, particularly in the presence of outliers, non-normal error distributions, and heteroskedasticity [14], [15].

B. Advantages Over OLS

In contrast to OLS, which assigns equal weight to all observations, robust regression reduces the influence of extreme values, resulting in more stable estimates when data are contaminated [16], [17]. Robust methods are especially effective with heavy-tailed or heteroskedastic error distributions [17], [18].

C. Types of Robust Regression

Common robust regression techniques include Huber regression, Tukey's bisquare regression, and M-estimators [15], [16]. Huber regression combines squared and absolute loss functions, while Tukey's bisquare further limits the influence of large residuals [15]. M-estimators generalise maximum likelihood estimation by using bounded influence functions [15], [19].

D. Comparison Between OLS and Robust Regression

OLS performs well under ideal assumptions but is highly sensitive to outliers and violations of those assumptions. In contrast, robust regression may sacrifice some efficiency under ideal conditions but offers greater stability and reliability with real-world data [16], [20]. Empirical results confirm that robust regression provides superior explanatory performance when classical assumptions are not met [20], [21].

E. Model Evaluation

Robust models are evaluated using adapted goodness-of-fit measures and diagnostic tools developed for robust estimators [16], [18]. Comparisons between full and reduced robust models offer insight into variable relevance while maintaining resistance to influential observations [20], [21].

IV. METHODOLOGY

A. Dataset Source

The dataset was obtained through an online survey conducted among undergraduate students at BINUS University. A total of 45 valid responses were collected, each containing information related to the respondent's monthly financial capacity, expenditure patterns, behavioural financial habits, and demographic characteristics. The unit of analysis is an individual student, and all monetary values are reported in Indonesian Rupiah (IDR).

B. Variable Definition

The dependent variable in this study is total monthly expenditure (y), measured in Indonesian Rupiah (IDR).

Ten predictors were included as independent variables. The first five variables represent quantitative spending and activity indicators, while the remaining five capture behavioural financial tendencies on a 1–5 Likert scale in Table 1.

Table 1 Definition of Variables Used in the Study

Code	Variable Description	Type
X_1	Monthly allowance (IDR)	Numeric
X_2	Living arrangement	Categorical (factor)
X_3	Frequency of eating out per week	Numeric
X_4	Frequency of leisure activities per week	Numeric
X_5	Total monthly transportation spending (IDR)	Numeric
X_6	Budgeting habit	Likert-scale numeric (1–5)
X_7	Expense-tracking habit	Likert-scale numeric (1–5)
X_8	Saving-before-spending habit	Likert-scale numeric (1–5)
X_9	Impulsive buying tendency	Likert-scale numeric (1–5)
X_{10}	Price-comparison habit	Likert-scale numeric (1–5)
Y	Total monthly expenditure	Numeric

Responses to the living arrangement variable (X_2) were originally offered through four survey options: living with parents/family, boarding/rented housing, dormitory, and others. These responses were recoded into four corresponding factor levels ("parent", "boarding", "dormitory", and "others") for analytical consistency. In the collected dataset, only the categories "parent" and "boarding" were observed, and therefore these two factor levels were used in subsequent modelling.

All numeric variables (X_1, X_3–X_{10}, y) were converted into numeric types prior to analysis to ensure compatibility with statistical procedures.

C. Data Preprocessing

Prior to statistical modelling, several preprocessing steps were performed to ensure data consistency and analytical validity. Survey responses were first cleaned using automatic header normalisation to remove irregular characters and standardise variable names. Numeric variables related to expenditure and behavioural indicators (X_1, X_3–X_{10}, and Y) were converted into numeric types to enable quantitative operations.

The categorical variable representing living arrangement (X_2) was recoded into four factor levels corresponding to the available response options in the questionnaire ("parent", "boarding", "dormitory", and "others"). Since only "parent" and "boarding" were observed in the collected dataset, these two levels were retained in the final modelling frame to avoid sparse factor representation. No missing values were present in the dataset, and therefore, no imputation procedures were required.

D. Assumption Diagnostics

Model assumptions were evaluated following standard OLS diagnostic procedures. Residual normality was tested using the Lilliefors version of the Kolmogorov-Smirnov test [11], [13]. Homoskedasticity was assessed using the studentized Breusch-Pagan test [13], and independence of residuals was examined using the Durbin-Watson statistic [13].

Multicollinearity was assessed using the Variance Inflation Factor (VIF), computed for each predictor. A threshold value of 10 was adopted to flag potential multicollinearity concerns [11]. Observations with potential influence on model estimation were identified using standardised residuals, leverage values derived from the hat matrix, and Cook's distance [22].

Assumption test results and the motivation for adopting robust regression are presented in the Results and Discussion section.

V. RESULT AND DISCUSSION

This section presents the empirical findings of the multiple regression analysis. First, the multiple linear regression model is estimated, followed by evaluation of classical assumptions to verify estimator validity. Next, a robust regression model is implemented to address the residual non-normality detected in the multiple linear regression model. A comparative model evaluation is conducted, and significant predictors are interpreted in the context of student expenditure behaviour.

A. Multiple Linear Regression Model Estimation

A multiple linear regression model was initially estimated using the Ordinary Least Squares (OLS) method with total monthly expenditure as the dependent variable and ten financial and behavioural predictors as independent variables. The estimated regression model is expressed as follows:

$$\hat{y} = 1280000 + 0.7148X_1 - 522600X_2(boarding) + 70410X_3 - 126700X_4 + 0.4228X_5 + 1.782X_6 - 15430X_7 - 308700X_8 + 46450X_9 + 63910X_{10}$$

The multiple linear regression model attained a residual standard error of 1,074,000, with R^2=0.62 and adjusted R^2=0.5082, indicating that approximately half of the variation in monthly expenditure is explained by the predictors. However, to ensure the estimator's validity, classical regression assumptions must be satisfied.

B. *Diagnostic Evaluation of Multiple Linear Regression Assumptions*

To ensure the validity of the OLS estimator, several classical regression assumptions were evaluated. For each diagnostic test, the null hypothesis assumes that the corresponding classical regression assumption holds, while the alternative hypothesis indicates a violation of the assumption. Hypothesis testing is conducted at a 5% significance level. The null hypothesis is rejected when the p-value is less than 0.05.

1) *Normality of Residuals*

The Lilliefors test was applied to evaluate residual normality. Results yielded p = 0.01608, indicating that the residuals deviate significantly from the normal distribution. This violation suggests that parameter estimates may be sensitive to outliers or non-Gaussian errors.

2) *Homoskedasticity*

Breusch-Pagan testing produced p = 0.8776, indicating no evidence of heteroskedasticity. Thus, the assumption of constant variance is satisfied.

3) *Non-Autocorrelation*

The Durbin-Watson test yielded a statistic of 1.651, with p = 0.117, providing no evidence of positive autocorrelation. Because the dataset comes from individual surveys rather than time series, independence is expected and confirmed.

4) *Multicollinearity*

Multicollinearity was evaluated using the Variance Inflation Factor (VIF). The resulting VIF values ranged from 1.25 to 3.36 (Table 2), which are substantially below the conventional threshold of 10. This indicates the absence of serious multicollinearity among predictors, supporting coefficient stability and justifying the retention of all variables in the full model.

Table 2 Variance Inflation Factor (VIF) Diagnostics

Predictor	VIF
X_1	1.554542
X_2	1.741894
X_3	1.887658
X_4	2.021920
X_5	1.266955
X_6	3.356462
X_7	3.148904
X_8	1.254987
X_9	1.351470
X_{10}	1.431801

5) *Outliers and Influential Observations*

Outlier and influence diagnostics were performed to assess observation-level effects on coefficient estimation. Standardised residual inspection identified one observation (ID 32) exceeding the |3| threshold, indicating an outlier in the response variable. Leverage diagnostics revealed three observations (IDs 3, 5, and 42) exceeding the recommended cutoff $\frac{2(k+1)}{n}$, suggesting disproportionately strong influence on fitted values.

However, no observations exceeded Cook's distance of 1, indicating that although several cases exert more leverage than average, none individually distort the regression fit to an extreme degree. Nonetheless, the combination of non-normal residuals and the presence of outlying/high-leverage observations implies that OLS coefficient estimates may not be fully robust to deviations from classical assumptions.

C. *Robust Model Estimation*

A robust regression model using the MM-estimator was implemented with the same predictor set. The estimated model is given by:

$$\hat{y} = -253700 + 0.7223X_1 + 174300X_2(boarding) + 17600X_3 + 72670X_4 + 0.5174X_5 - 93280X_6 + 86280X_7 - 32340X_8 + 13830X_9 + 5964X_{10}$$

The robust model achieved R^2=0.882 and adjusted R^2=0.847, substantially increasing explanatory power relative to the OLS estimator. Additionally, the residual standard error decreased from 1,074,000 (OLS) to 516,200, indicating improved fit and reduced sensitivity to outliers.

D. *Comparative Model Evaluation*

Table 3 Model Performance Comparison

Model	R-squared	Adjusted R-squared	Residual Standard Error
OLS	0.62	0.508	1,074,000
Robust Regression	0.882	0.847	516,200

Based on Table 3, the robust model provides significantly better explanatory performance. This improvement aligns with the theory stating that robust regression yields more reliable estimators under non-normal residuals and outlier influence.

E. *Interpretation of Predictors*

To evaluate the statistical significance of each predictor in the robust regression model, individual hypothesis tests were conducted at the 5% significance level. For each predictor X_j, the hypotheses are defined as follows:

H_0: $\beta_j = 0$ (the predictor has no effect on total monthly expenditure)

$H_1: \beta_j \neq 0$ (the predictor has a significant effect on total monthly expenditure)

where β_j denotes the regression coefficient associated with predictor X_j., and the null hypothesis is rejected when the corresponding p-value is less than the chosen significance level (α=0.05).

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Table 4 Robust Regression Coefficients and Significance

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	-253700	763400	-0.332	0.742
X_1	0.7223	0.2654	2.721	0.010
X_2(boarding)	174300	299200	0.583	0.564
X_3	17660	85340	0.207	0.837
X_4	72670	102500	0.709	0.483
X_5	0.5174	0.2938	1.761	0.087
X_6	-93280	114300	-0.816	0.420
X_7	86280	116000	0.744	0.462
X_8	-32340	130400	-0.248	0.806
X_9	13830	77760	0.178	0.860
X_{10}	5964	103300	0.058	0.954

Table 4 summarises the estimated coefficients and statistical significance of all predictors under the robust regression model. Based on the hypothesis testing results, only the monthly allowance (X_1) is statistically significant at the 5% level and positively associated with total monthly expenditure, indicating that students with greater income capacity tend to spend more overall. Meanwhile, transportation cost (X_5) shows a marginally significant positive relationship ($p \approx 0.087$), suggesting that mobility-related expenses may contribute to increased spending, but evidence remains insufficient at the conventional 5% threshold.

Other predictors, including consumption frequency variables (X_3–X_4), budgeting behaviours (X_6–X_{10}), and living arrangement (X_2), exhibit no statistically significant association with total expenditure under the robust estimator. Although the coefficient signs vary, these variables do not provide meaningful explanatory power for predicting monthly spending in the observed sample.

Overall, the results highlight that financial capacity plays the dominant role in shaping student expenditure, while lifestyle frequency and behavioural self-regulation factors do not translate into measurable differences in total monthly spending within this dataset.

VI. Conclusion

From the research that has been conducted, it can be concluded that the Robust Regression model produces a more stable and accurate model than the Ordinary Least Squares model in analysing student spending factors. This statement is evidenced by a significant increase in the R^2 value from 0.62 to 0.882, followed by a decrease in the residual standard error from 1,074,000 to 516,200. This indicates that the robust model is more effective in analysing data that has outliers and is not normally distributed.

The test results also found that the Monthly allowance (X_1) is the only factor that has a statistically significant influence on total student spending. Meanwhile, other variables such as Total monthly transportation spending (X_5) only show marginal significance, and lifestyle factors and financial habits (such as saving or budgeting) do not have a significant measurable impact in this study.

Overall, these findings confirm that financial capacity or income remains a major determinant of student spending behaviour, and the use of robust statistical methods is crucial to ensure the reliability of findings when the classical assumptions of regression are not met.

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